論文翻訳 4007 佐藤駿一

論文の翻訳手順

- とりあえず複数のLLMで日本語をリファイン
    - 授業で使った重複表現の削除、表現の具体化、最小主義、動詞中心など...
    - 専門用語を抜き出し、それの英訳を手動で行う
- 複数LLMを使って英訳

プロンプト

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# タスク
あなたはプロの技術文書の翻訳家です。以下の制約条件と入力文に従って、最高の翻訳を出力してください。


# 制約条件
* 入力された日本語の文章を、明確（clear）、簡潔（concise）、そしてフォーマルな技術英語（formal technical English）に翻訳してください。
* 翻訳時には、以下の「日本語原文のリファイン方針」と「翻訳品質向上のための指針」を最大限に考慮し、反映させてください。
* 出力形式は「出力形式」に従って出力してください


## 日本語原文のリファイン方針（翻訳時に考慮すること）
1. **重複表現の削除**:

    * 例：「リチャージャブルバッテリーの充電」のような表現は、文脈に応じて「バッテリーの充電」または「リチャージャブルバッテリー」のように、より簡潔な表現にしてください。

2. **曖昧な表現の具体化**:

    * 例：「できるだけ頻繁に」のような曖昧な頻度を表す表現は、可能であれば「1日1回以上」のように具体的な表現に置き換えることを意識してください。数値表現（例：「数千万」）も、可能であればより具体的な数値（例：「3600万」）を原文が示唆している場合はそれに従ってください。原文に具体的な情報がない場合は、その曖昧さを保ちつつも明確さが損なわれないように翻訳してください。

3. **実質的な情報量が少ない表現の見直し**:

    * 例：「～ということが重要です」「～することが期待される」といった、実質的な情報量が少ない、あるいは冗長な表現は、より直接的で簡潔な表現に置き換えてください。

4. **名詞構文の動詞化**:

    * 例：「修正を行う」は「修正する」、「説明である」は「説明する」のように、名詞的な表現を動詞中心の表現に変換してください。



## 翻訳品質向上のための指針
1. **動詞中心の表現**:

    * 名詞を多用した複雑な文章よりも、シンプルな動詞を使用した文章にしてください。例:「be contaminated」→「polluted」

2. **能動態の優先**:
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* 原則として能動態（active voice）を使用してください。受動態（passive voice）の使用が適切、または自然である場合に限り、受動態を使用しても構いません。

3． **学術的・技術的な文脈の維持**：

* 翻訳対象が学術的な文章または技術的な内容を含む場合、その文脈にふさわしい専門用語や表現スタイルを選択してください。

4． **前置詞・冠詞の用法**

* 原文のニュアンスにあった前置詞・冠詞を使用してください。前置詞や冠詞を使用した際には謎その全つぃや冠詞を使用したものかを考えてください。

5． **用語の統一**

* 専門用語や固有名詞は全文を通して統一してください。また、正しい表現を心がけてください。

出力形式

1． **掲載雑誌**

* 電気系学会の学会誌に掲載する形式にしてください

2． **出力レベル**

* Analytical, PhD-levelな出力してください

入力文
[ここに翻訳したい日本語の文章を入力してください]

出力文

- LLMどおしの比較
  - それぞれの英訳を自分とLLMで読んで批評、それぞれの良い場所をLLMを使って抽出
  - 評価点
    - 文章の統一性
    - 文章の可読性
    - 用語の統一性
    - あいまいな表現を避け、具体的か
    - 動詞中心の表現
    - 能動態の優先
    - 前置詞･冠詞の用法

- 使用モデル
  - ChatGPT o3
  - Claude 3.7 sonnet
  - Genimi 2.5 pro

英訳:

1. Introduction

A significant issue for urban railways in metropolitan areas is delays during the morning rush hour. From approximately 6:00 AM to 8:00 AM, railways experience the highest passenger volume of the day, prompting operators to implement various measures to increase transport capacity. Among these, shortening train headways is a crucial measure as it directly impacts capacity and significantly affects passenger convenience. However, while shortening headways improves capacity, it also exacerbates the effects of delays. Under current conditions, resolving a delay once it occurs is time-consuming, leading to operational adjustments such as train cancellations and headway regulations to restore spacing between trains. Consequently, the operator-defined transport capacity is not maintained, and passengers experience longer travel times, leading to decreased satisfaction. Therefore, there is a demand for achieving operations with minimal delays while maintaining transport capacity.

Meanwhile, when implementing countermeasures for delays, their impact is typically assessed beforehand using simulators. In conventional train operation simulators, simulations are run by inputting basic data such as route and vehicle information, along with station dwell times. The dwell times used for this input are generally based on actual train operation data (hereafter referred to as "train operation performance data"). This makes it impossible to input the altered dwell times that would result from changes in route geometry or timetables implemented as delay countermeasures.

Therefore, we posited that if station departure delay times could be predicted using machine learning based on train operation performance data, it would become possible to determine departure delays based on changes in travel times resulting from route modifications or changes in station handling. This would enable the verification of the effectiveness of delay countermeasures within a train operation simulator. This paper describes a method for predicting train departure delay times, intended for integration into a train operation simulator, and evaluates its prediction accuracy.

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2. Background and Objectives

2.1 Research Background

Many conventional railway lines operate on a fixed-block signaling system, which ensures safety by restricting train speeds over predetermined distances (1)(2). This system limits the speed of a following train as it approaches a preceding train and the interval between them shortens. Due to this mechanism, when headways are reduced to increase transport capacity, a delay in a preceding train's departure from a station has a greater impact on the following train than it would with wider headways. During off-peak hours, headways are sufficiently long, so minor delays at stations do not significantly disrupt the timetable. However, during congested periods, headways are short, causing even small delays to affect subsequent trains and propagate without being resolved. This propagated delay continues to escalate and often remains unresolved even after the rush hour has passed, posing a significant problem.

When implementing countermeasures for such delays, their effectiveness is verified in advance using simulators. Currently, while it is possible to input modified route data for a countermeasure, the potential delays that might occur at stations cannot be predicted. Therefore, the impact is evaluated by inputting existing train operation performance data. However, changes to route geometry or station layout alter the influence of a preceding train on a following train, meaning departure delays will not be the same as in the existing data. Consequently, a quantitative evaluation of delay countermeasures using existing performance data is difficult, necessitating a method to estimate delays occurring at stations from operational data.

2.2 Related Research

In recent years, research on train delay prediction has become more active in Japan. However, much of this research aims to support traffic management operations, and few studies have focused on chronic delay propagation as a key feature.

As a foundational study in train operation prediction, Kunimatsu et al. proposed a method for estimating passenger numbers and predicting delays using a train operation and passenger behavior simulator. However, these estimations use regression equations derived from statistics of known data and do not reflect the daily variations in delays, thus failing to capture the characteristics of delay propagation (3). In research using machine learning for train delay prediction, Tatsui et al. proposed a method using Long Short-Term Memory (LSTM), which is known to be effective for time-series data prediction, and reported an accuracy of 38.9% within a 5-second error margin (4)(8). Furthermore, Iwamoto et al. developed a predictive timetable function that forecasts delays in the near future and visualizes them on a diagram, which has been implemented in the operation management systems of some railway lines. However, this requires users to set numerous parameters according to the station, train type, and time of day, and it cannot predict delay times based solely on train movements (5). Moreover, in these studies, the input data for the predictor is limited to stations behind the target station, failing to capture the characteristics of propagating delays.

Meanwhile, delay prediction research is also being conducted overseas. A study by Zhou et al. used a technique called stacking, which employs multiple weak learners, to predict train delay times on railways in Northeast China. This prediction achieved an accuracy of 90% within a 30-second margin. However, the target lines are high-speed railways with wide train headways, which differs significantly from the complex delay situations in Japan's metropolitan areas (6).

2.3 Position of this Research

The objective of this study is to enable a more accurate verification and evaluation of the effectiveness of delay countermeasures using a train operation simulator. To achieve this, it is necessary to learn the complex relationship between train running and station delays from train operation performance data using machine learning, predict delay times, and compare the station-generated delays that would change before and after a countermeasure is implemented. Previous research used data from stations behind the target station as input for the learner to predict delay times. However, since chronic delays propagate from ahead of the train, the characteristics of these delays were not being learned. Therefore, in this paper, we constructed a Convolutional Neural Network (CNN) (7) model that can learn the features of delay propagation by including the running data of preceding trains from sections ahead of the target station, which is known at the time the target train arrives at the target station.

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3. Analysis of Train Operation Performance Data

3.1 Overview of Train Operation Performance Data

Train operation performance data is a record of all train movements on a given day, documenting the "train name, train type, arrival time at each station, departure time at each station, and arrival track." Arrival and departure times are recorded in seconds, providing high temporal resolution. In this paper, we calculated metrics such as arrival and departure delay times, inter-station running times, and others by comparing this data with the train timetable, and used these as training data. The calculated data, all in seconds, are listed below:

1. **Departure Delay Time** = Actual Departure Time - Scheduled Departure Time
2. **Arrival Delay Time** = Actual Arrival Time - Scheduled Arrival Time
3. **Inter-station Running Time** = Station Arrival Time - Previous Station Departure Time
4. **Delay Generated Between Stations** = Arrival Delay Time - Previous Station Departure Delay Time
5. **Delay Generated at Station** = Departure Delay Time - Arrival Delay Time
6. **Dwell Time** = Actual Departure Time - Actual Arrival Time
7. **Arrival Headway** = Actual Arrival Time - Preceding Train's Departure Time

3.2 Characteristics of Delays

Delays can be categorized into three states depending on the time of day. The graph in Fig. 1, which plots departure delay time (vertical axis) against time (horizontal axis) for a given station over a year, shows the increasing trend of delays. From Fig. 1, the transition of delays reveals three distinct periods:

1. A period where initial delays occur and begin to increase.
2. A period where the delay has peaked and stagnates.
3. A period where the accumulated delay begins to decrease.

Before period (1), various delays occur, but they do not show a tendency to escalate. However, upon entering period (1), a clear trend of escalating delays is observed. This period corresponds to the time when passenger volume increases and train headways narrow, making it easier for dwell times to exceed the scheduled duration, thus amplifying delays. In period (2), the accumulated delay remains constant without further increase. This is a period where operational adjustments, such as headway regulation, are made to control the spread of delays. In period (3), the previously constant delay time shows a decreasing trend. This period corresponds to a time when passenger volume decreases and train headways widen. During this time, delay-control measures combined with wider headways allow for the rapid resolution of accumulated delays.

Fig. 2 shows a train diagram illustrating the delays occurring during period (1). The blue line represents the trajectory if the train ran according to the scheduled timetable, while the red line represents the actual train trajectory. Therefore, the discrepancy between the blue and red lines indicates the delay time. Fig. 2 confirms that when a delay occurs on one train, it slows down the following train, causing the delay to propagate.

The focus of this paper is the propagating delay shown in Fig. 2, which occurs during period (1).

3.3 Inter-station Correlation of Delays Generated at Stations

If the delay time at one station correlates with the delay times at other stations, it might be possible to estimate the potential delay at a station without using machine learning. Therefore, we conducted a correlation analysis of the departure delay times at each station to check for relationships. Fig. 3 shows a heatmap visualizing the extent of delay influence. Correlation coefficients are generally interpreted as "almost no correlation" for 0-0.2, "slight correlation" for 0.2-0.4, "moderate correlation" for 0.4-0.7, and "strong correlation" for 0.7-1.0.

In Fig. 3, the horizontal axis represents trains and the vertical axis represents stations. The data at origin (0,0) is the reference train and station for comparison. The color, from red to blue, indicates the strength of the correlation with the reference data. For example, the data at (1,1) represents the train following the reference train at the next station, and its color shows the correlation with the data at (0,0). Trains travel from A to G. The data at (0,0) is compared with itself, so it has the strongest correlation of 1.0. The meaning of the data in each quadrant of Fig. 3 is as follows:

- **First Quadrant:** Future data at stations behind the target station.
- **Second Quadrant:** Past data at stations behind the target station.
- **Third Quadrant:** Past data at stations ahead of the target station.
- **Fourth Quadrant:** Future data at stations ahead of the target station.

Fig. 3 reveals that the range of data correlating with the delay at a given station has distinct characteristics. Analyzing the influence of delays on a station by quadrant: The first quadrant confirms that a delay at one station propagates to subsequent trains at following stations, with a large area showing "moderate correlation." The second quadrant, which represents the input data used in previous research, shows few data points with "moderate correlation." Specifically, for the third preceding train, the correlation at the first station behind the target station is only "slight," indicating a weak relationship with the generated delay. In contrast, the third quadrant shows that for the third preceding train, the correlation at the first station ahead of the target station is "moderate." This pattern holds for the first and second preceding trains as well, suggesting that the delay at a given station is influenced by preceding trains at stations ahead. The fourth quadrant shows no significant correlation. This is a reasonable result given the characteristics of delay propagation described in Section

3.1. Since delays propagate when a delay on one train affects the running of the following train, the relationship with subsequent trains at stations ahead (fourth quadrant) is weak.

From this analysis, it is suggested that including data from the third quadrant, in addition to the second quadrant, may be effective for predicting the delay time generated at a station.

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4. Delay Prediction with a Machine Learning Network

4.1 Network Architecture and Input Data

The network used in this paper is a Convolutional Neural Network (CNN). While many studies on train delay prediction use LSTM, we hypothesize that delay propagation is not an event confined to individual trains but is influenced by preceding trains, as shown in Section 3.3. Therefore, we treated the data as discrete and adopted a CNN, which can learn spatial relationships between individual data points. To learn the features of delay propagation, the data from the section of the line that preceding trains have already traveled is more important than the data from the section the target train has just traveled. In reality, there are sections where the preceding train has not yet arrived at a forward station by the time the target train arrives at its station. However, since the purpose of this research is to evaluate delay countermeasures on a simulator, we created the input data assuming that all data from preceding trains is available.

The network used in this paper is shown in Fig. 4. The baseline network (hereafter, Model 1) is a CNN model composed of three 3D convolutional layers and four fully connected layers, with input data limited to stations behind the target station. We constructed three additional networks to compare their accuracy against Model 1, using data from both behind and ahead of the target station. The differences between the created models and their input data are shown in Table 1. The explanatory variables for Models 2 to 4 include data from stations ahead of the target station. Only Model 2 is constructed with LSTM. Models 3 and 4 are both constructed with CNN, but Model 3 is the model where parts of the forward station data that are unknown at the time the target train arrives at the station are filled with zeros, while Model 4 is the model that assumes all information about preceding trains is known and uses it as input. For all models, the loss function used was mean absolute error (mae), and the optimization algorithm was adam.

4.2 Data Used

The delay targeted in this study is one that gradually increases. Therefore, we do not target large, sudden delays. Such delay times are considered learning noise, so only data falling within the ranges shown in Table 2 were used.

The data used consists of a total of 1600 data points from trains operating between 7:00 AM and 9:00 AM on weekdays in 2019, after removing noise. Of this, 80% was used as training data and 20% as test data to compare prediction accuracy. The test data was not continuous but was selected randomly.

4.3 Accuracy with Each Input

We compare the prediction accuracy for six stations on a line passing through a major metropolitan area. At each of the target stations on this line, there is no overtaking or waiting for rapid trains; all trains stop. The time period considered is the morning rush hour, when train headways are the shortest of the day.

To evaluate prediction accuracy, we compared the percentage of predictions with an absolute error between the actual and predicted values of 5 seconds or less. The accuracy for each model is shown in Table 3. Comparing the baseline Model 1 and the LSTM-based Model 2, the accuracy of the CNN-based Model 1 was more than 10 percentage points higher than Model 2 for 5 out of the 7 stations. For the remaining 2 stations, the accuracy was also about 7 points better. The lower accuracy of Model 2 compared to the other models suggests that delay propagation is not an event occurring in individual train units but is influenced by preceding and following trains.

Since the CNN model was confirmed to be able to learn the relationships of delays between trains, we examine the differences in input explanatory variables. First, comparing Model 1 and Model 3, an improvement in accuracy is seen in 3 of the 7 stations, with a maximum improvement of about 6 percentage points. However, at other stations, the accuracy worsened, particularly at stations E and beyond, where one station showed a 4-point decline. The difference between stations B, C, and D, where accuracy improved, and stations E, F, and G, where it worsened, may be the presence or absence of connections to other lines. Since our model's explanatory variables do not include congestion rates, it does not account for fluctuations in passenger numbers due to transfer timings. Therefore, at stations where congestion might be a more significant cause of delay than preceding trains, inputting information from forward stations may have acted as noise. Furthermore, Model 3 pads unknown information from preceding trains with zeros, which may have amplified the effect of noise. Comparing Model 3 with Model 4, which assumes all information about preceding trains is known, the prediction accuracy worsens at only one station, while the others show similar or improved results. Moreover, the stations where accuracy improved were E, F, and G, and at station G, an improvement was seen even compared to Model 1. This strongly suggests that the information from forward stations that is unknown upon arrival was important.

Comparing Models 2 to 4, it was confirmed that Model 4, which assumes all information about preceding trains is known, has the highest accuracy. However, compared to Model 1, there are stations where accuracy did not improve. Therefore, we compared the prediction accuracy of Model 1 and Model 4 while changing the tolerated absolute difference between actual and predicted values to 3 seconds, 5 seconds, and 10 seconds. The results are shown in Table 4. From Table 4, it can be confirmed that for stations B, C, and D, the superiority in prediction accuracy does not change even when the tolerance is altered. Therefore, it can be said that information from forward stations contributes to prediction accuracy. On the other hand, for stations E and F, where Model 4's accuracy was worse than Model 1's, the prediction accuracy improves depending on the tolerated absolute value. This suggests that the network configuration and hyperparameter tuning may not be sufficient, and there is a possibility that accuracy could be improved at these stations. However, at station F, since the prediction accuracy is worse than Model 1's for all tolerance levels, it can be said that the information from forward stations is acting as noise.

From the above, we have shown that CNN can capture the characteristics of propagating delays and may be effective for train delay prediction. We also clarified that data from ahead of the target train's target station is an effective explanatory variable for delay prediction.

4.4 Prediction Accuracy of Model 4

Fig. 5 shows the results of using Model 4 to predict the station-generated delay during the morning rush hour at Station G on a certain day and calculating the station departure delay time. In Fig. 5, the vertical axis is the station departure delay time, and the horizontal axis represents trains. The blue line is the actual delay time for that day, and the orange line is the predicted value.

Overall, it can be seen that the model is able to predict the entire process from the increase to the decrease of the delay. In particular, it was confirmed that the phenomenon of increasing delays during the rush hour is accurately predicted. On the other hand, the prediction accuracy is poor in the parts where the delay stagnates and then decreases. This is because the training data did not include data from outside the period of increasing delays. Therefore, by including a wider time range in the training data, there is a possibility of accurately predicting not only the time when delays increase but other periods as well.

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5. Conclusion

In this paper, with the goal of developing a train operation simulator for the quantitative evaluation of pre-implementation delay countermeasures, we proposed a machine learning model and input data structure for predicting train station-generated delays. We compared the delay prediction accuracy for small, cumulative delays that occur during the morning rush hour. The results showed that, with a tolerated absolute difference of 5 seconds between actual and predicted values, the CNN achieved an average prediction accuracy of about 50%, demonstrating that CNN can capture the features of delay propagation more effectively than LSTM. Furthermore, by adding data from stations ahead of the target station as an explanatory variable, the prediction accuracy improved by up to 10 percentage points, confirming that the proposed method of constructing explanatory variables based on the characteristics of delay propagation is effective for delay prediction.

The network in this paper evaluated the prediction accuracy using only time-based data obtained from train operation performance data. However, the fact that prediction accuracy did not improve at some stations even with the proposed method indicates the need to input information that contributes more to delays, such as congestion rates. In the future, it will be necessary to clarify the relationship between congestion rates and delays and to construct a delay prediction method that takes congestion rates into account.